Mini Project Report

On

"SPENDWISE – SMART EXPENSE TRACKER"

Submitted To

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Submitted By

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CERTIFICATE

This is to certify that following student:

Ms. VAIBHAVI A (Roll NO- 20231CSE0712)

of Third Year (CSE) has successfully completed their work entitled **"SPENDWISE – SMART EXPENSE TRACKER".**

Bikram Sarkar
Assistant Professor
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Presidency University

ACKNOWLEDGEMENT

I would like to express my profound gratitude to **Bikram Sarkar, Assistant Professor,** and for their contributions to the completion of Mini project titled **SPENDWISE – SMART EXPENSE TRACKER**.

I would like to express my special thanks to our mentor Mr. Bikram Sarkar for his time and efforts he/she provided throughout the Semester. Your useful advice and suggestions were really helpful to us during the project's completion. In this aspect, I am eternally grateful to you.

I would like to acknowledge that this project was completed entirely by me and not by someone else.

Ms. VAIBHAVI A

(Roll NO-20231CSE0712)

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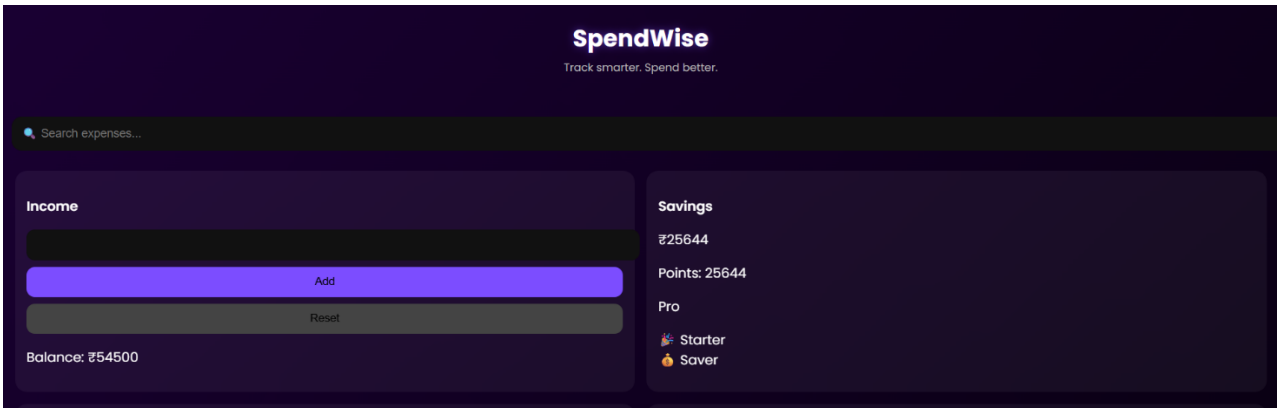
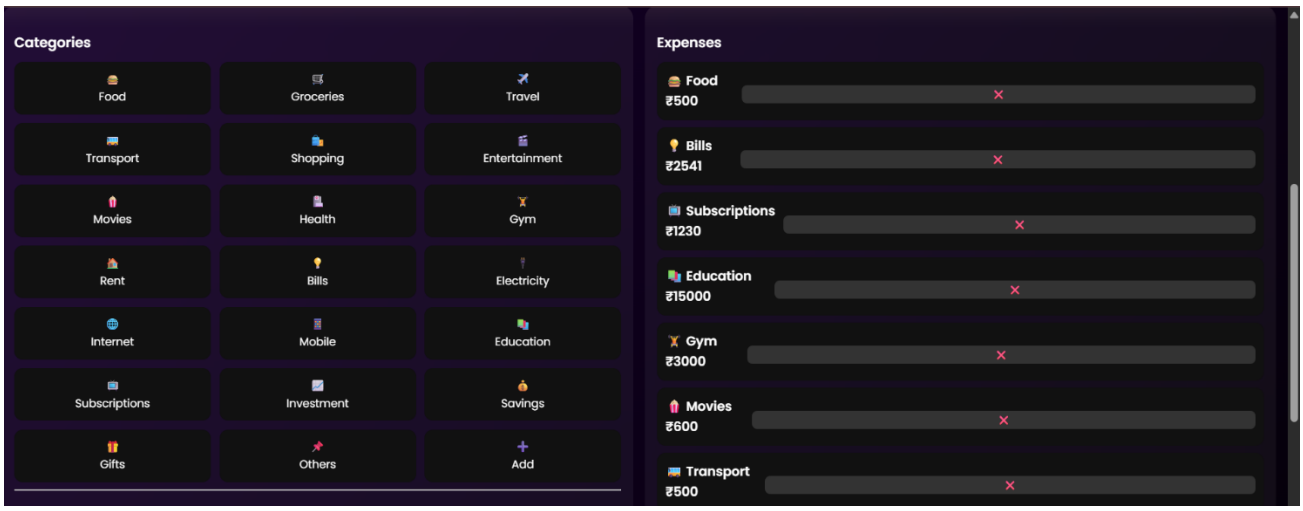
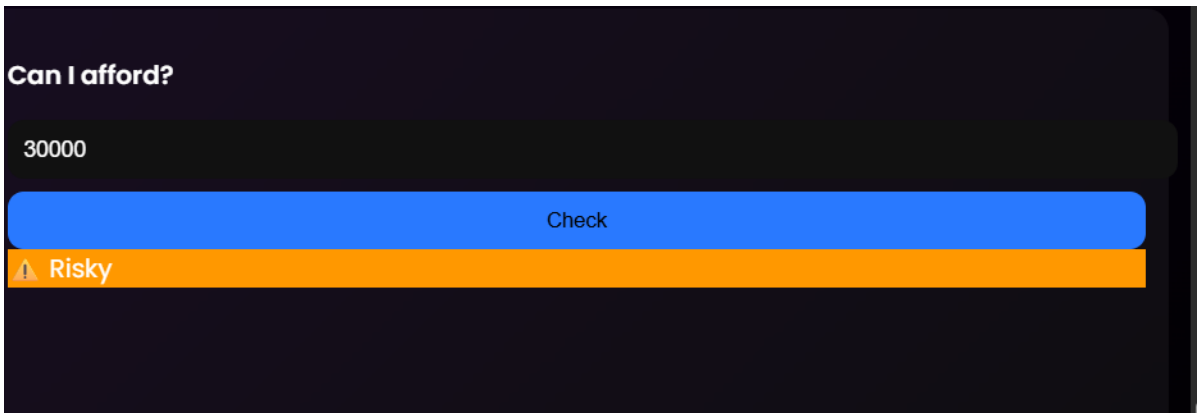
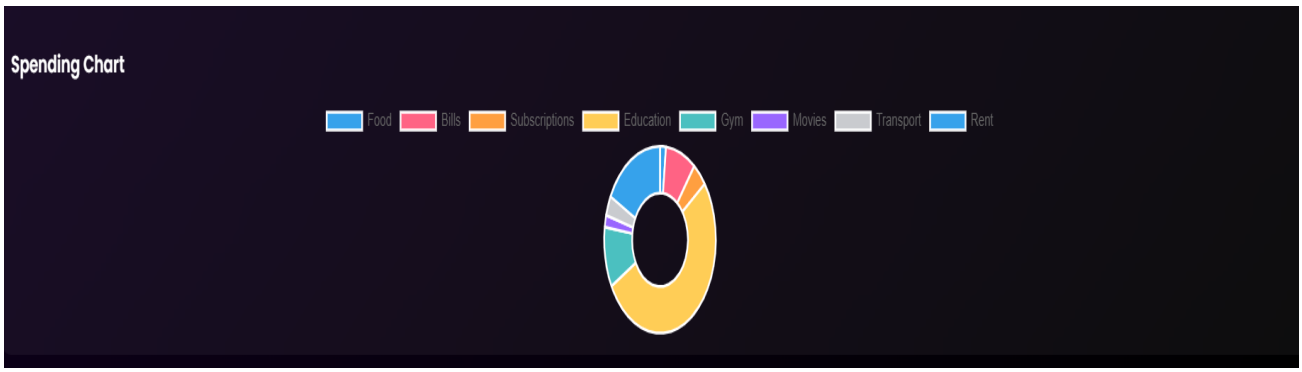
I. LIST OF IMAGES & FIGURES





Can I afford?
8497
Check
Safe


Can I afford?
30000
Check
Risky


Spending Chart
Food
Bills
Subscriptions
Education
Gym
Movies
Transport
Rent

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II. ABSTRACT

The **SpendWise Smart Expense Tracker** is a web-based application developed to provide a simple, efficient, and user-friendly platform for managing personal finances. In today's fast-paced digital world, tracking income and expenses has become essential for maintaining financial stability, and this project aims to simplify that process using modern web technologies such as HTML, CSS, and JavaScript.

The system is designed with a clean, modern, and visually appealing user interface featuring a dark-themed layout with glass morphism effects. It allows users to add income, record expenses, and categorize their spending into predefined categories such as Food, Travel, Shopping, Bills, and more. Users can also create custom categories, making the system flexible and adaptable to individual needs. Each expense is stored with details such as amount, category, date, and icon, providing a structured overview of financial activities.

The application includes essential features such as real-time expense tracking, savings calculation, and an interactive search functionality to filter expenses quickly. It also integrates a dynamic doughnut chart using Chart.js to visually represent spending patterns across different categories. Additionally, the system includes a financial goal tracker that helps users set savings targets and monitor their progress through a visual progress bar.

To enhance user experience, the project includes a unique "Can I afford?" feature, which analyzes the user's financial status and provides instant feedback on whether a particular expense is safe, risky, or unaffordable. All data is stored using the browser's localStorage, ensuring persistence without requiring a database or backend server. The application is fully responsive and works seamlessly across mobile, tablet, and desktop devices, making it an ideal beginner-friendly project that demonstrates the effective use of frontend technologies and client-side data management.

1. INTRODUCTION:

With the increasing use of digital tools and smart devices, managing personal finances has become both essential and challenging. Many individuals struggle to track their income, expenses, and savings effectively, which often leads to poor financial planning. To address this issue, digital expense tracking applications have gained popularity as they provide users with a simple and efficient way to monitor their financial activities.

The **Smart Expense Tracker "SpendWise"** is developed as a web-based application designed to help users manage their finances in an organized and user-friendly manner. The main objective of this project is to create a platform where users can track their income, categorize their expenses, and analyze their spending habits easily. The application features a modern and visually appealing interface with components such as a dashboard, category section, expense list, savings tracker, and graphical representation of spending patterns.

The system includes predefined categories such as Food, Travel, Shopping, Bills, and Entertainment, along with the option for users to add custom categories. Each expense entry includes details like amount, category, date, and visual icons, making the data easy to understand. The application also provides additional functionalities such as savings calculation, goal tracking, affordability checking, and a dynamic chart to visualize spending distribution.

This project is developed using HTML and CSS for designing the frontend interface, JavaScript for implementing interactive functionalities and data handling, and browser local storage for saving user data. The SpendWise application not only demonstrates the working of a personal finance management system but also provides practical knowledge of building responsive and interactive web applications.

OBJECTIVES

The main objective of the **SpendWise – Smart Expense Tracker** is to develop a simple and user-friendly web application that helps users manage their personal finances effectively. The system allows users to record income and expenses, enabling better tracking and control over financial activities.

The project aims to categorize expenses into predefined and custom categories, helping users understand their spending patterns clearly. It also provides real-time calculations of total income, expenses, and savings, allowing users to make informed financial decisions.

Another objective is to present financial data through visual representations such as charts, making analysis easier and more interactive. The application also includes features like goal tracking and affordability checking to encourage better saving habits and financial discipline.

Additionally, the system ensures data persistence using browser local storage so that user data is localStorage and accessible even after refreshing the application.

Finally, the project provides practical experience in developing interactive web applications using HTML, CSS, and JavaScript, demonstrating how technology can be used to solve real-life financial management problems efficiently.

3. Literature Review

Several studies and projects have explored the development of personal finance and expense tracking systems using modern web technologies. The following points summarize the key findings from existing research and similar applications:

3.1 Web-Based Financial Management Systems

Personal finance management systems have gained significant attention in recent years due to the increasing need for budgeting and expense tracking. Earlier systems were manual or spreadsheet-based, while modern applications use web technologies to provide automated tracking and real-time insights. Technologies like HTML, CSS, and JavaScript are commonly used to build lightweight, interactive financial tools suitable for students and individuals.

3.2 JavaScript and Local Storage for Data Management

JavaScript provides powerful features for handling dynamic data on the client side. The use of **localStorage** allows applications to store user data directly in the browser without requiring a backend database. Studies show that local storage is an efficient solution for small-scale applications like expense trackers, as it ensures data persistence and faster performance without server dependency.

3.3 Category-Based Expense Tracking Systems

Expense tracking applications commonly use category-based organization to help users analyse their spending habits. Research highlights that categorizing expenses (such as Food, Travel, Bills, etc.) improves financial awareness and decision-making. Many systems also allow custom category creation to provide flexibility and personalization for users.

3.4 Data Visualization in Financial Applications

Visualization techniques such as charts and graphs are widely used in financial applications to represent spending patterns. Libraries like Chart.js enable developers to create interactive and responsive charts. Studies indicate that visual representation of data helps users better understand trends and manage their finances more effectively.

3.5 Responsive and User-Centered Design

Modern web applications focus on responsive design to ensure usability across devices. With a large percentage of users accessing applications via mobile devices, techniques like CSS Grid, Flexbox, and media queries are essential. Research emphasizes that a clean UI and intuitive design significantly enhance user experience and engagement.

4. Project Details

4.1 Technology Stack

The following technologies were used to develop the SpendWise Smart Expense Tracker:

Technology	Version	Purpose in Project
HTML	HTML5	Page structure, forms, layout, UI components
CSS	CSS3 + Fonts	Styling, responsive design, modern UI effects
JavaScript	ES6	Core logic, event handling, dynamic updates
Chart.js	Latest CDN	Data visualization (expense charts)
Local Storage	Browser API	Stores income, expenses, categories, and goals
VS Code	Latest	Development environment

4.2 Project Structure & File Description

The project follows a simple and organized structure:

```
SpendWise/
├── index.html      → Main Application Page
├── style.css       → UI Styling and Layout
├── script.js       → Core Logic and Functionality
├── assets/         → Icons (if used)
└── README.md       → Project Description
```

4.3 Data Structure (Local Storage)

All user data is stored in the browser using localStorage, avoiding the need for a database.

Data stored includes:

- income → Total income added by user
- expenses → Array of expense objects
- categories → Default + custom categories
- goal → Savings goal details
- points → Gamification score

Example structure:

```
expenses = [
  {amount: 200, category: "Food", date: "2026-04-15"}
]
```

4.4 Core Functionality — JavaScript Implementation

4.4.1 Data Persistence

Local storage is used to store all financial data. Functions like localStorage.setItem() and getItem() ensure data is retained even after page reload.

4.4.2 Income & Expense Management

Users can:

- Add income
- Add expenses with category and date
- Delete expenses

The system automatically calculates:

- Total expenses
- Remaining savings

4.4.3 Category Management

- 20 default categories (Food, Travel, Bills, etc.)
- Users can add custom categories
- Each category is assigned an icon dynamically

4.4.4 Data Visualization

- Implemented using Chart.js
- Displays category-wise expense distribution
- Doughnut chart updates dynamically

4.4.5 Goal Tracking & Affordability Check

- Users can set savings goals
- Progress bar shows completion percentage
- “Can I afford?” feature evaluates spending decisions based on savings

4.4.6 Gamification Features

- Points based on savings
- Levels such as Beginner, Saver, Pro
- Badges to motivate users

4.5 Frontend — HTML & CSS Design

Sub-section	Description
4.5.1 Design Philosophy	Modern dark theme with purple gradient for a premium feel
4.5.2 Typography	Google Font "Poppins" used for clean readability
4.5.3 Layout System	CSS Grid for responsive card-based layout
4.5.4 Responsive Design	Media queries for mobile and desktop compatibility
4.5.5 UI Components	Cards, buttons, inputs with glass morphism effect
4.5.6 Visual Effects	Blur, shadows, hover animations for interactive UI

5. How to Run the Project

5.1 Prerequisites

- A modern web browser (Google Chrome, Firefox, or Edge)
- A code editor (VS Code recommended)
- Basic understanding of HTML, CSS, and JavaScript
- The SpendWise project folder (containing index.html, style.css, script.js)

5.2 Steps

- Download or extract the **SpendWise project folder**
- Open the folder in VS Code or any code editor
- Locate the file **index.html**
- Right-click on **index.html** and select **"Open with Live Server"**
- The application will open in your browser
- Start using the app:
- Add income
- Select or create categories
- Add expenses
- View savings and charts

5.3 Important Notes

- No server or database setup is required the project runs completely in the browser
- All data is stored using **localStorage**, so it remains saved even after refreshing
- Clearing browser cache or local storage will remove all saved data
- The application works offline once loaded in the browser
- Charts are generated using **Chart.js CDN**, so internet is required only for initial loading

6.Advantages

- No database or server setup required runs directly in the browser using localStorage
- Simple and user-friendly interface suitable for beginners
- Modern, responsive UI with dark theme and smooth visual effects
- Real-time tracking of income, expenses, and saving
- Category-based expense management improves financial awareness
- Users can create custom categories for personalized tracking
- Interactive charts (Chart.js) provide clear visualization of spending patterns
- Goal tracking feature helps users achieve financial targets
- "Can I afford?" feature assists in smart spending decisions
- Gamification system (points, levels, badges) encourages saving habits
- Data persists even after refreshing the page (using localStorage)
- Works offline after initial load no internet required for core features
- Beginner-friendly code with clear logic and structure
- Easily customizable and extendable for future improvements
- Can be deployed on any static hosting platform (GitHub Pages, Netlify)

7. Limitations

• Uses browser localStorage, which is not suitable for large-scale or multi-user applications

• Data is stored only on the client side and can be lost if browser cache or storage is cleared

• No user authentication system cannot support multiple users or accounts

• No backend or database integration (MySQL, Firebase, etc.) for permanent data storage

• Limited data security since information is stored in the browser

• Chart.js requires internet for initial loading (CDN dependency)

• No cloud sync data cannot be accessed across different devices

• Not suitable for enterprise-level financial management

• No export options (PDF/Excel) for financial reports

• Limited scalability due to purely frontend implementation

• No advanced analytics or AI-based financial insights

• Performance may slow down if a large number of expenses are stored

8. Conclusion and Future Scope

8.1 Conclusion

The **SpendWise Smart Expense Tracker** successfully demonstrates the development of a modern, interactive web application using core frontend technologies HTML, CSS, and JavaScript. The project provides a complete solution for personal finance management, allowing users to add income, track expenses, categorize spending, and visualize financial data through charts.

Key concepts demonstrated in this project include dynamic data handling using JavaScript, data persistence through browser localStorage, responsive UI design using CSS Grid and media queries, and interactive data visualization using Chart.js. The application also incorporates additional features such as goal tracking, affordability checking, and basic gamification elements, making it more engaging and practical for users.

The project runs entirely in the browser without requiring any backend or database setup, making it lightweight, easy to use, and convenient for demonstration purposes. Overall, SpendWise effectively showcases how web technologies can be used to solve real-life problems like personal financial management in a simple and efficient manner.

8.2 Future Scope

- Integrate a backend (Node.js / PHP) with a database (MySQL / MongoDB) for secure and scalable data storage
- Implement user authentication system for multi-user support
- Add cloud synchronization to access data across multiple devices
- Introduce export features (PDF, Excel) for financial reports
- Integrate AI-based insights for smart budgeting and expense prediction
- Add reminders and notifications for bill payments and savings goals
- Improve data security using encryption techniques
- Develop a mobile application version (Android / iOS)
- Add advanced analytics such as monthly trends and comparisons
- Enable multi-currency support for global usability
- Deploy the application on cloud platforms (Netlify, Vercel, or AWS)

9. References

1. W3Schools – HTML, CSS, and JavaScript Tutorials

   https://www.w3schools.com

2. MDN Web Docs – JavaScript, Web APIs, and Local Storage

   https://developer.mozilla.org

3. Chart.js Official Documentation

   https://www.chartjs.org/docs/latest/

4. Google Fonts – Typography Resources

   https://fonts.google.com

5. Stack Overflow – Developer Community Discussions

   https://stackoverflow.com

6. JavaScript.info – Modern JavaScript Tutorial

   https://javascript.info

7. CSS-Tricks – Guides on CSS Grid, Flexbox, and Responsive Design

   https://css-tricks.com

8. FreeCodeCamp – Web Development Tutorials and Projects

   https://www.freecodecamp.org

10. Appendix — Key Code Snippets

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SpendWise</title>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<!-- PREMIUM FONT -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700&display=swap" rel="stylesheet">

<style>
body {
  margin:0;
  font-family:'Poppins', sans-serif;
  background:linear-gradient(135deg,#1a0033,#000);
  color:white;
}

/*  HEADER */
.header {
  padding:20px 15px;
  text-align:center;
}
.header h1 {
  font-size:28px;
  font-weight:700;
  margin:0;
  text-shadow:0 0 10px rgba(124,77,255,0.6);
}
.header p {
  font-size:13px;
  color:#bbb;
  margin-top:4px;
}

/* SEARCH */
.search-bar {padding:10px;}
.search-bar input {
```

```
.card {
  background:rgba(255,255,255,0.05);
  padding:14px;
  border-radius:14px;
  backdrop-filter:blur(10px);
}

/* HEADINGS */
.card h3 {
  font-size:16px;
  font-weight:600;
  margin-bottom:10px;
}

/* INPUT */
input {
  width:100%;
  padding:10px;
  margin-top:6px;
  border-radius:10px;
  border:none;
  background:#111;
  color:white;
  font-size:14px;
}

/* BUTTON */
button {
  width:100%;
  padding:10px;
  margin-top:8px;
  border:none;
  border-radius:10px;
  cursor:pointer;
  font-weight:500;
}

.green {background:#7c4dff;}
.red {background:#ff4081;}
.blue {background:#2979ff;}
.gray {background:#444;}
```

```
  background:#7c4dff;
  width:0%;
}

@media(max-width:768px){
  .container {grid-template-columns:1fr;}
}
</style>
</head>

<body>

<div class="header">
  <h1>SpendWise</h1>
  <p>Track smarter. Spend better.</p>
</div>

<div class="search-bar">
<input id="search" placeholder="🔍 Search expenses..." oninput="renderExpenses()">
</div>

<div class="container">

<div class="card">
<h3>Income</h3>
<input id="incomeInput" type="number">
<button class="green" onclick="addIncome()">Add</button>
<button class="gray" onclick="resetApp()">Reset</button>
<p>Balance: ₹<span id="incomeDisplay">0</span></p>
</div>

<div class="card">
<h3>Savings</h3>
<p>₹<span id="savings">0</span></p>
<p>Points: <span id="points">0</span></p>
<p id="level"></p>
<div id="badges"></div>
</div>

<div class="card">
<h3>Categories</h3>
```

```
<script>

/* 20 DEFAULT CATEGORIES */
const defaultCategories = [
{name:"Food",icon:"🍔"},
{name:"Groceries",icon:"🛒"},
{name:"Travel",icon:"✈️"},
{name:"Transport",icon:"🚌"},
{name:"Shopping",icon:"🛍️"},
{name:"Entertainment",icon:"🎬"},
{name:"Movies",icon:"🎥"},
{name:"Health",icon:"🩺"},
{name:"Gym",icon:"🏋️"},
{name:"Rent",icon:"🏠"},
{name:"Bills",icon:"💡"},
{name:"Electricity",icon:"🔌"},
{name:"Internet",icon:"🌐"},
{name:"Mobile",icon:"📱"},
{name:"Education",icon:"📚"},
{name:"Subscriptions",icon:"📺"},
{name:"Investment",icon:"📈"},
{name:"Savings",icon:"💰"},
{name:"Gifts",icon:"🎁"},
{name:"Others",icon:"📌"}
];

function getSafeCategories(){
  try{
    let data = JSON.parse(localStorage.getItem("categories"));
    if(!data || !Array.isArray(data) || data.length === 0){
      localStorage.setItem("categories", JSON.stringify(defaultCategories));
      return [...defaultCategories];
    }
    return data;
  }catch{
    localStorage.setItem("categories", JSON.stringify(defaultCategories));
```

```
if(points>500) badges+="🎯 Starter<br>";
if(points>2000) badges+="💰 Saver<br>";

document.getElementById("badges").innerHTML=badges;
save();
}

function checkAfford(){
let amt=Number(document.getElementById("affordInput").value);
let total=expenses.reduce((s,e)=>s+e.amount,0);
let savings=income-total;
let result=document.getElementById("affordResult");

if(!amt) return;

if(amt<=savings){
result.innerHTML="✅ Safe";
result.style.background="#00c853";
}else if(amt<=savings*1.5){
result.innerHTML="⚠️ Risky";
result.style.background="#ff9800";
}else{
result.innerHTML="❌ Not affordable";
result.style.background="#ff5252";
}
}

function resetApp(){
if(confirm("Reset everything?")){
localStorage.clear();
location.reload();
}
}

renderCategories();
updateUI();

</script>

</body>
</html>
```

GITHUB LINK: